

sphere encased by a cylinder is two-thirds that of the cylinder—to be his greatest achievement, and he asked for the shapes to be carved on his tombstone.

Royal assignment

The Roman writer Vitruvius recorded a possibly apocryphal story of how one of Archimedes' most important discoveries was made. King Hieron II suspected that his new crown was not pure gold, so he asked his chief scientist to investigate. Having weighed the crown, Archimedes realized that if he also knew the crown's volume, he could work out its density—and since metals have different densities, he could prove whether it was pure gold.

Archimedes pondered how to measure the volume of irregular objects on a visit to the public bath house. As he got into the tub, he noticed the water rising and spilling over the side, and he felt lighter. He is said to have shouted “*Eureka!*” (“I found it!”) and run home naked. He had realized that the volume of water spilled from the bath was equal to his body's volume, and that an object immersed in a liquid receives an upward force equal to the weight of the liquid it displaces.

In practice, the effect of water spilled by the crown would have been small and difficult to measure. Instead, Archimedes may have measured the upthrust by hanging the crown and an equal weight of gold on both ends of a horizontal stick, which he then lowered into water. If the crown was pure gold, both objects would receive an equal upthrust and the stick would stay level.

Military mechanics

“Archimedes' principle,” as it came to be known, was also put to militaristic use. Archimedes used it to design

Syracusia, a 2,000-man ship given as a gift to Ptolemy III, and devised a system of levers and pulleys for launching the vessel. The royal fleet also benefited from the Archimedes screw, a device that he may have seen in Egypt for pumping water up from the bilges of ships. When Syracuse was besieged by the Romans in 213 BCE, Archimedes invented the Claw of Archimedes—a mechanical hook designed to lift an enemy ship by the prow, throwing sailors into the sea and capsizing or damaging the vessel. Despite orders that he should not be harmed, he was killed by a Roman soldier in 212 BCE.

“Give me a **lever** and a **place to stand ... and I'll move the Earth.**”

Archimedes

Archimedes' findings on measuring volume and the buoyancy of objects in a fluid proved that Hieron II's crown was not pure gold, but an alloy of gold and other, less dense metals.

